

- 5 An intrinsic semiconductor has a valence band that is completely filled and a conduction band that is empty at  $T = 0$  K. There are no mobile charge carriers available for conduction of electricity.

An intrinsic semiconductor has a small energy band gap between the bottom of the conduction band and the top of the valence band. When temperature rises, electrons in the valence band are thermally excited and gain sufficient energy to reach the conduction band, leaving behind holes in the valence band.

Both the electrons in the conduction band and the holes in the valence band are mobile charge carriers. When an external electric field is applied, the mobile charge carriers contribute to the flow of electric current.

The increase in number of mobile charge carriers enables a larger current with the same potential difference. Therefore the electrical resistance of the intrinsic semiconductor material decreases.

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| Note: Effect of holes in valence band and the part they play in conduction must be described. |
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